International Migrants & Refugee Data

Auditing, modelling, and querying a two-source UN dataset on global migration and displacement

Project Overview

This project is an SQL exercise built around a real United Nations migration dataset, combining International Migrant Stock figures (UN Population Division) with refugee and asylum data (UNHCR). The dataset covers 263 areas: a mix of individual countries, regional groupings, continents, and a world total.

The goal was to audit the data for structural problems first, design a schema that accounts for those problems, and then move on to analysis. As detailed below, this dataset has several genuine dataquality issues that shaped every later query, most notably that it interleaves two separate reporting systems within the same table, with no flag distinguishing them.

Tools

- Database engine: SQLite, via DB Browser for SQLite (DB4S)
- Source file: SYB68_327_202511_International Migrants and Refugees.csv (~6966 rows)
- Techniques covered: data auditing, schema design, joins, CTEs, window functions, correlated subqueries

Data Audit Findings

Before writing analytical queries, the dataset was audited for structure, completeness, and integrity. This stage surfaced four issues that affected every query written afterward.

Long data format

The CSV, by default, was saved in long format. The fix was to first load the data in power query and pivot the attribute and value columns. This compacted the data into wide format where the attributes are inverted to categorical columns used for the analysis.

Two interleaved source systems, not one

The single biggest structural finding: each row is not simply keyed by Area + Year, but by Area + Year + Source. The same area and year can appear twice: once from the UN Population Division (which reports migrant stock numbers) and once from UNHCR (which reports refugee, asylum-seeker, and other-of-concern figures). Each source only populates its own subset of columns; the remaining columns are blank for that row.

This was confirmed directly against the data:

```
SELECT t04, region_country_area, year, source, COUNT(*) AS no

FROM Migrants

GROUP BY t04, region_country_area, year, source

ORDER BY no DESC;
```

Duplicate UNHCR submissions for the same area, year, and source

A closer look at the duplicate-key investigation above revealed that for a number of area/year combinations (for example, Albania 2023, Algeria 2015 and 2023), the UNHCR source appears twice or thrice. These are not exactly duplicates. They are split submissions, where one row carries the refugee and asylum-seeker counts and a second row carries only the “other of concern” figure for the same area, year, and source label. The “Total population of concern to UNHCR" also appears to be the sum of refugee, asylum-seeker and “other of concern” values but in certain instances the “Total population of concern to UNHCR" values are either null or the sum isn’t exact.

Ninety-three such duplicate area+year+source combinations were found in the raw file. Left unresolved, a careless SELECT * or join would visually appear to duplicate data; the correct handling is to aggregate these rows together so each metric is captured exactly once per area-year-source key and add a new total UNHCR column.

```
CREATE TEMPORARY TABLE migrants_temp AS SELECT
        t04,
        region_country_area,
        year,       source,
        ims_both_sexes_4num,
        ims_both_sexes_4pop,
        ims_male_4pop,
        ims_female_4pop,
        NULLIF(SUM(COALESCE(`refugees_total_4num`, 0)), 0) AS `refugees_total_4num`,
        NULLIF(SUM(COALESCE(asylum_seekers_total_4num, 0)),0) AS asylum_seekers_total_4num,
```

```
        NULLIF(SUM(COALESCE(other_of_concern_total_4num, 0)),0) AS other_of_conc                AS
        NULLIF(SUM(COALESCE(total_pop_of_concern_unhcr_4num,                 0)),0)
total_pop_of_concern_unhcr_4num,
        NULLIF(
                SUM(COALESCE(`refugees_total_4num`, 0))
                + SUM(COALESCE(asylum_seekers_total_4num, 0))
                + SUM(COALESCE(other_of_concern_total_4num, 0))
        ,0) AS unhr_calc_total
FROM Migrants

GROUP BY t04, region_country_area, year, source

ORDER BY t04, region_country_area, year, source;
```

Countries, regions, continents, and a world total in one column

The Region/Country/Area column mixes genuine countries with regional aggregates (e.g. Western Africa, Southern Asia), continents (Africa, Asia, Europe), and a single “Total, all countries or areas” row, with no flag column distinguishing the three. A query that sums migrant stock across this column without filtering would double- or triple-count, since continent totals already include their constituent countries’ figures.

This was resolved with a manually built lookup table mapping each T04 code to a country name and continent, used throughout the project to isolate country-level rows from aggregate.

Completeness summary

Null rates across the key metric columns, computed once the two-source structure was understood:

Column	Rows	Nulls	% Null
migrant_stock_number	1,912	862	45.1%
migrant_stock_pct_of_population	1,912	862	45.1%
refugees_total	1,912	1,190	62.2%
asylum_seekers_total	1,912	1,214	63.5%
other_of_concern_total	1,912	1,332	69.7%
total_population_of_concern	1,912	1,146	59.9%

Table 1. Null rates per metric column. High null rates here reflect the two-source row structure, not poor data quality: each row only ever populates the columns belonging to its own source.

Schema Design

Based on the audit findings above, the raw import was not queried directly. Instead, a cleaned working table (migrants_temp) and a small supporting lookup table were built first.

migrants_temp

Built by grouping the raw import on (t04, region_country_area, year, source) and aggregating the UNHCR metric columns. This collapses the 93 split UNHCR submissions into one row per true areayear-source combination, while preserving NULL as a meaningful “not reported” signal rather than overwriting it with zero.

lookup

A small manually-built reference table mapping each t04 code to a country name and continent. Its purpose is purely to let queries cleanly separate country-level rows from continent/region/worldaggregate rows.

Analysis

Each section below states the business question, the key SQL technique used to answer it, the result, and a brief takeaway.

Top 10 countries by migrant stock (latest available year)

Technique: window function (ROW_NUMBER) to select the correct row per country, avoiding the column-mismatch trap of an ungrouped aggregate.

Country	Migrant stock	Year
United States of America	52,375,047	2024
Germany	16,750,084	2024
Saudi Arabia	13,683,841	2024
United Kingdom	11,845,479	2024
France	9,186,757	2024
Spain	8,870,527	2024
Canada	8,805,839	2024
Country	Migrant stock	Year
United Arab Emirates	8,157,000	2024

Australia	8,111,404	2024
Russian Federation	7,605,774	2024

Table 2. Top 10 countries by international migrant stock, most recent reporting year per country (UN Population Division source).

Takeaway: the United States holds more than three times the migrant stock of the next-highest country (Germany), and the top 10 is dominated by high-income destination economies in North America, Western Europe, and the Gulf.

Migrant stock as % of population, ranked within continent

Technique: DENSE_RANK() partitioned by continent, ordered by year and percentage, to reveal the top countries within each continent rather than a single global ranking that would be dominated by one region.

Ranking within continent (rather than globally) matters here specifically because migrant stock as a share of population is heavily skewed by a small number of city-states and Gulf economies; a global top-10 would show almost nothing about typical patterns within, say, Africa or Asia. Partitioning by continent surfaces the leading countries in every region, not just the global outliers.

Takeaway: small, high-income, labor-importing economies (Gulf states, financial hubs) consistently top their respective continental rankings, while the percentage figures for most large, populous countries remain in the low single digits. This points to economic pull factors, not population size, as the driver of migrant stock concentration.

The most noteworthy country is the Holy See (Vatican City) in Europe where 100% of the population are migrants. This reveals the fact that the Holy See has no permanent citizen.

Total refugees by continent, most recent reporting year

Technique: filtering to lookup rows where country IS NULL and continent IS NOT NULL, isolating UNHCR's own continent-level aggregate rows rather than re-summing from country rows (which avoids double-counting).

Continent	Refugees	Most recent year
Asia	12,201,435	2024
Continent	Refugees	Most recent year
Europe	9,816,951	2024
Africa	8,776,888	2024
Northern America	420,131	2019

Latin America & the Caribbean	220,962	2019
Oceania	41,860	2024

Table 3. UNHCR-reported refugee totals by continent, most recent available year per continent.

Takeaway: Asia, Europe, and Africa each carry refugee populations in the 8–12 million range, dwarfing the Americas and Oceania. Critically, the table also exposes a coverage gap: Northern America and Latin America & the Caribbean have no UNHCR continent-level figure more recent than 2019, while the other four continents report through 2024.

Pivoting both sources into a single row per area-year (CTE)

Technique: two CTEs filtering migrants_temp by source, joined back together with LEFT JOIN on (t04, year).

```
WITH UNPD AS (
        SELECT *
        FROM migrants_temp
        WHERE ims_both_sexes_4num IS NOT NULL
        ),
        UNHCR AS (
        SELECT *
        FROM migrants_temp
        WHERE refugees_total_4num IS NOT NULL
        )
        SELECT
                u1.t04,
                u1.region_country_area,
                u1.year,
                u1.ims_both_sexes_4num,
                u1.ims_both_sexes_4pop,
u1.ims_female_4pop,             u1.ims_male_4pop,
        u2.refugees_total_4num,
u2.asylum_seekers_total_4num,
u2.other_of_concern_total_4num,
u2.total_pop_of_concern_unhcr_4num
```

```
FROM UNPD as u1

LEFT JOIN UNHCR AS u2

ON u1.t04 = u2.t04

AND u1.year = u2.year;
```

Takeaway: this query resolves the dataset's core structural problem by reconstructing one logical row per area-year with both migrant stock and refugee figures side by side.

Countries with the highest refugee-to-migrant-stock ratio

Technique: self-join on migrants_temp using t04, year, and source < source as a deduplication trick to pair each country's UNHCR row with its UN Population Division row exactly once.

Takeaway: this ratio reveals countries where displacement pressure is large relative to total migration stock, typically countries that are major refugee-hosting or refugee-origin states (e.g. Iran, Chad, Ethiopia) rather than traditional migration destinations (e.g. USA, Germany, UAE).

Countries appearing in one source but never the other

Technique: two CTEs of distinct area codes per source, combined with a LEFT JOIN ... WHERE right.code IS NULL anti-join pattern in both directions, unioned together.

Takeaway: the areas found only in UN Population Division data are predominantly small territories and micro-states (e.g. Andorra, Bermuda, Greenland, San Marino, various Pacific island territories) that fall outside UNHCR's reporting scope, given that UNHCR's mandate focuses on refugeeproducing and refugee-hosting situations rather than universal population coverage.

Limitations & Caveats

Some limitations are inherent to the dataset itself and apply to any analysis built on top of it, regardless of how carefully the SQL is written:

1. Non-continuous time series. Years are not evenly spaced (5-year gaps early, annual later, a 2year gap at the end, with 2022 missing entirely). Any "change over time" metric must specify which years are actually being compared.
2. Inconsistent reference years across continents. As shown in Table 3, two of six continents have no UNHCR figure more recent than 2019, while the rest report through 2024.

Appendix: Dataset Reference

Source columns

Raw column	Cleaned name	Source system
International migrant stock: Both sexes (number)	ims_both_sexes_4num	UN Population Division
International migrant stock: Both sexes (% pop.)	ims_both_sexes_4pop	UN Population Division
International migrant stock: Male (% pop.)	ims_male_4pop	UN Population Division
International migrant stock: Female (% pop.)	ims_female_4pop	UN Population Division
Total refugees and people in refugee-like situations	refugees_total_4num	UNHCR
Asylum seekers, including pending cases	asylum_seekers_total_4num	UNHCR
Other of concern to UNHCR	other_of_concern_total_4num	UNHCR
Total population of concern to UNHCR	total_pop_of_concern_unhcr_4num	UNHCR

Table 4. Mapping from raw CSV headers to the cleaned column names used in migrants_temp.